REFERENCE

1. *Industrial Ventilitation Manual*, 17th edition, American Conference of Governmental Industrial Hygienists, Cincinnati, 1982.

3

Basic Hood Design

If you were going to design and build a successful automobile you would first establish all the requirements: safety, economy, reliability, maintenance, load capacity, cost and so on. There is no use in specifying a 350-hp engine and putting in a midget drive train; the wrong alloy in the frame and it may crack; a plastic body that may warp or spark plugs that cannot be replaced.

How do you avoid these problems? You do so by understanding what each component does and then start making judgment calls as to what you want and how you are going to achieve it. We are not designing or building cars, there are enough people in that market. In this book we are concerned about building fume hoods. To do so we must set up a checklist of our requirements and be sure that what we want is what is built. Anything less than the safest and best is just not acceptable.

We reasonably understand automobiles because we are constantly made aware of what makes them tick. However, in the world of fume hoods we are not quite so fortunate. There are some good published fume hood articles and papers, and there are some very poor ones. How do you judge the good and the bad? Now we will start our bare bones analysis of the fume hood and see how all the various parts really fit together. You will then be in a better position to make the judgment calls on the various articles yourself and with a pretty good level of confidence.

For purposes of clarification, and to be sure we are all calling the same part by the same name, let's review accepted fume hood nomenclature as shown in Figure 3.1.

SIZE

The size of a hood, say, 6 ft, refers to the outside (side to side) dimension of 6 ft, or very close to it. The interior work chamber will be 6 ft less the thicknesses of the two side walls, which range from $3\frac{1}{2}$ to 6 in. per side. The depth of the hood again is measured from the outside shell and can vary from roughly 32 to 37 in. depending on the manufacturer and on the particular model or type. The depth of the interior work chamber is primarily affected by the air foil design and back baffle design and is partially influenced by the laboratory workbench depth.

Why worry about size? First it is easy to make the mistake that a 6 ft hood has 6 ft of interior work space. Secondly, the architect/engineer many times selects the size based on allocated floor space. To me that is the tail wagging the dog. The laboratory must have hoods compatible with user

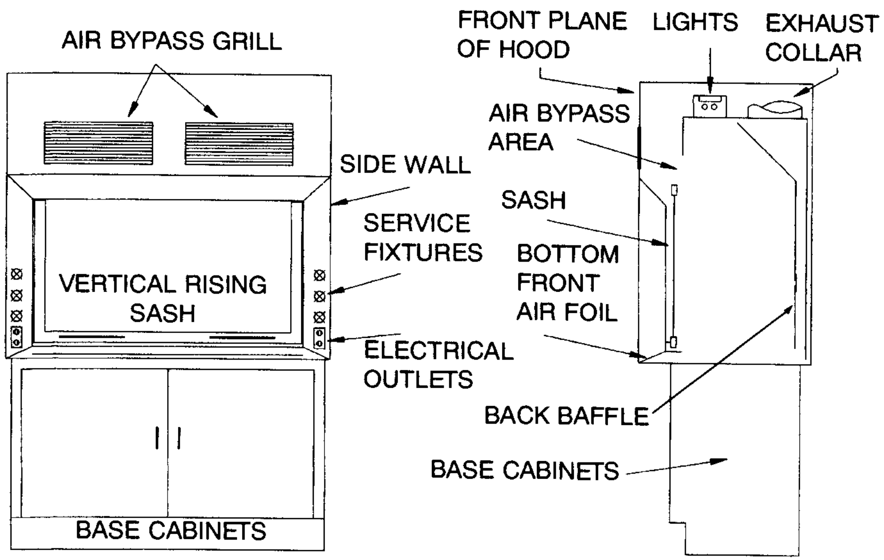

FIGURE 3.1 *Fume hood nomenclature.*

requirements. If a 5 ft hood is sufficient, that's great. However, if you really need an 8 ft hood for your equipment and you wind up with a 5 ft hood, you have a real problem. Tell the planners what you need and somehow they will figure out how to provide it for you. I am firmly convinced that if you do a really good job of educating the architect/engineer as to your true and viable requirements, you will get what you need. Try a snow job and it always comes back to haunt you.

The more basic questions we should be asking, however, are not those of overall size but what is required to provide maximum performance: then see how much space it consumes.

SIDE WALL CONSTRUCTION

Let us consider the hood side walls. They serve two purposes: (1) to provide ample space for the mechanical and electrical services (i.e., air, gas, vacuum, 120 volts, etc.) and (2) to provide the dimensional detail for the aerodynamically shaped entrance to the hood chamber.

In considering the performance improvement that is supplied to the hood due to the side wall, smaller is not better. Some manufacturers can provide as small as a $3\frac{1}{2}$ in. thickness; however, most prefer to furnish a $4\frac{1}{2}$ to 6 in. thickness. The plumber really appreciates as much space as he or she can get so as to run the piping and your maintenance people will bless you for a larger size should repairs be required. All that aside, the real reason to consider a maximum thickness is that the thicker the wall, the more gentle the continuous slope of the aerodynamically shaped entrance. The more gentle the slope the less turbulence is caused by the vertical front edges of the opening. The less turbulence, the better the hood performance, refer to Figure 3.2.

My recommendation is for a $4\frac{1}{2}$ to 5 in. dimension with a continuous slope. This has a noticeable effect on improving hood performance.

HOOD DEPTH

While the side-to-side dimension of the hood is critical for equipment placement and use, the front-to-back size is more critical and less forgiving from a safety standpoint.

Depending on the manufacturer this overall dimension ranges from 32

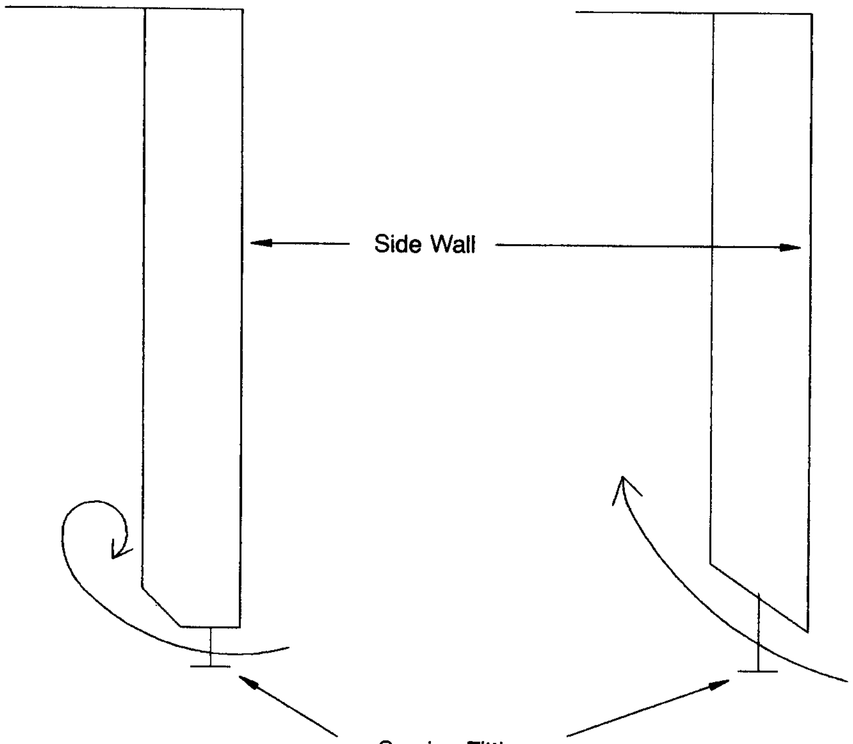

FIGURE 3.2 *Hood side wall design.*

to 37 in. Architects also have some influence here when they do the laboratory layout. Depending on workbench depth and aisle space there may be a demand to keep the hood depth as small as possible. This is not an area that should be compromised. First, you jeopardize safety and then you get caught up with too little space in which to work. So let's design the depth and include all those critical items that make up this front-to-back profile. The six segments of this part of the fume hood are outlined below and shown in Figure 3.3.

Quickly, let us consider our working surface allowance. If the hood has an overall depth of 37 in. the safe usable work space will be approximately 21 in. Dimensions A, C, and D (Figure 3.3) are most critical for superior hood performance. Each of these design parameters will be explored in detail later in this chapter.

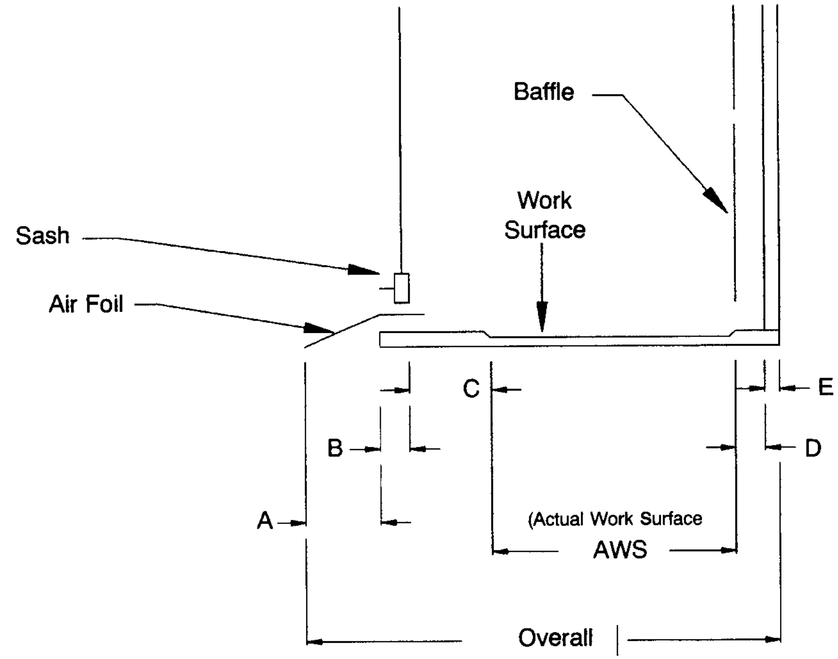

FIGURE 3.3 *Six critical design points for fume hood. (A) Air foil depth ($4\frac{1}{2}$ to 5 in.). (B) Sash thickness and set back from front edge of hood chamber, depends on both the sash and sash track construction (1 to $1\frac{1}{2}$ in.). (C) Recommended minimum work-free zone behind the sash for maximum safety (6 in.). (D) Air (duct) space between the back of the baffle and the hood lining (minimum of 2 in.). (E) Allowance for framing system, material thicknesses, and wall clearance 1 to $1\frac{1}{2}$ in.*

INTERIOR CHAMBER DIMENSIONS

The interior vertical size can be about any height that your procedures dictate. The standard manufactured dimension is approximately 48 in.; past history has sustained this height as being adequate for most operations. Manufacturers like it because it conforms to the 48-in. width of the uncut sheets of most hood lining materials. Their savings in materials are reflected in their pricing. Add any amount of height and the cost increase may seem disproportionately high. You must remember that the manufacturer has to make a special liner, an entirely special outside shell and baffle system, and maybe even a special sash. If you plan to have the interior height larger than 60 in. you should consider walk-in or distillation hood designs, since your back baffle system could become self-defeating due to improper slot design (refer to pages 17 to 23 and 41).

SASH OPENING SIZE

What other factors should we consider in discussing the height of the fume hood superstructure? For starters how about clear sash opening. Hood fabricators make their products in a manner most convenient to their manufacturing processes and tooling. By reviewing Figure 3.4 we shall arrive at some viable user requirements.

If: A = 72 in.
then: B = approximately 5 in. less than A.
C = 29 to 30 in.
D = 37 to 38 in.

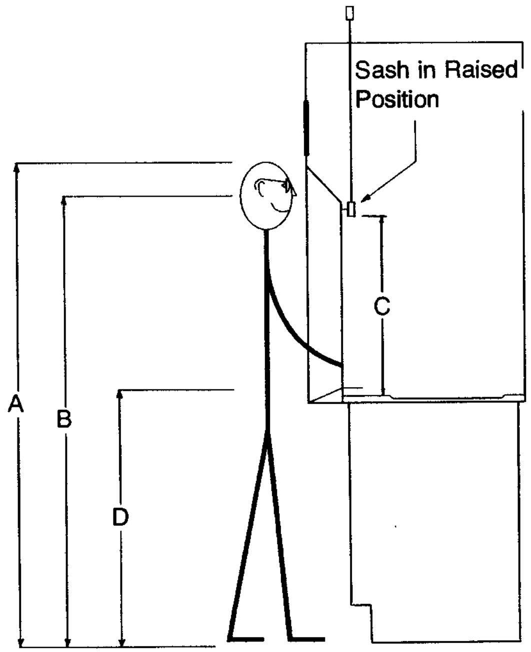

FIGURE 3.4 *User requirements regarding sash opening sizing.*

Dimension C is the opening of the sash. It should be at least 29 to 30 in. If you compare this opening size in various hood catalogs you will see that the clear opening size varies from 26 to 31 in. People are getting taller with each passing year, and I feel you should insist on at least a 31 in. clear opening; I have seen some rather large facilities insist on 33 in. If you have a large quantity of hoods the price difference between 31 and 33 in. is small. The added inch or two does not adversely affect the operational safety of the hood proper.

We have considered macro design; now let us down-size to our individual hood components.

AIR FOIL DESIGN

A fume hood that does not incorporate the air foil entrance shape is roughly 50 years behind the times. A square faced hood is an accident that is happening all the time even under pretty ideal conditions. Given a laboratory with some noticeable room air movement or personnel traffic problems, you can have a real disaster safety-wise when hoods lack the air foil design.

The most important section of the aerodynamic entrance shape is the bottom front air foil. The reason is very simple and easy to demonstrate (Figure 3.5).

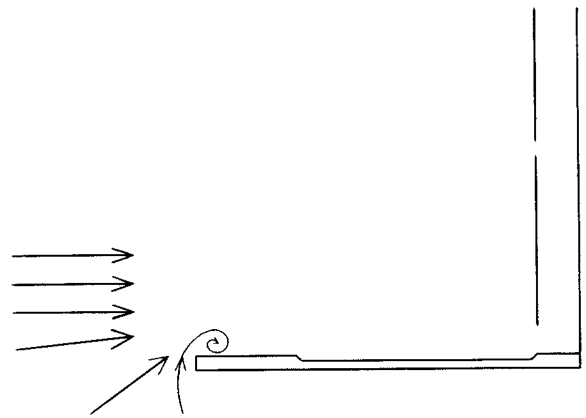

FIGURE 3.5 *Front edge turbulance without bottom front air foil.*

Remember one of the statements in the introduction was that "air is lazy." As air goes into a hood it arrives with both a horizontal and a vertical vector component, and these collide at the entrance to the hood at the work surface area. This forms a large roll that sits at the front edge of the hood and can (and does) bring hood contaminants from the interior of the hood to the front. If someone walks past the hood or there is a room cross-draft, your hood-contaminated air is being mixed with the room air.

This is easy to demonstrate. Take a pie pan and fill it with $\frac{1}{2}$ in. of warm water and place it about 3" back from the face of the hood. Break up some dry ice and put several reasonably small pieces in the water and see what happens to the CO_2 and water vapor [1]. A hood with no air foil has the turbulence well defined. Add an air foil of the proper design and "poof" the turbulent pattern just goes away (Figure 3.6). As repeated in many advertisements, "try it, you'll like it" or "don't leave home without one."

Now let us explore air foil design. Are they all created equal? Not really.

The slope of the foil should be reasonably gentle, say, an angle of 20° (Figure 3.7). Thirty degrees is no disaster, but you can get into trouble at 45°, especially if the spacing under the foil is greater than the normal 1 in.

Dimension A is pretty standard at a nominal 1 in. (1 in. less the metal thickness). Individual laboratory requirements due to electric plugs, tubing, and so on, where you want these to exit the hood under the foil, may require a spacing of $1\frac{1}{4}$ in. This is no real problem as long as dimension B is large enough to straighten out the air flow. When the space exceeds $1\frac{1}{4}$ in., you enter a problem area and could have a roll of turbulent air at the front edge under the foil.

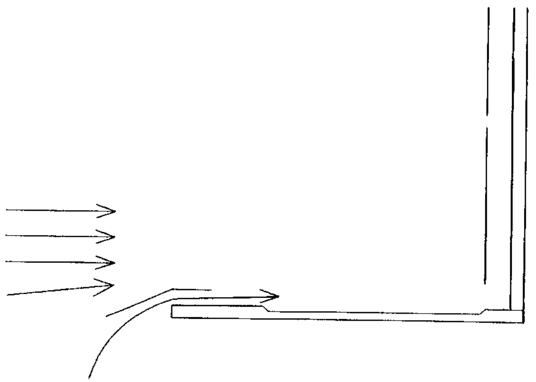

FIGURE 3.6 *No turbulance with bottom front air foil.*

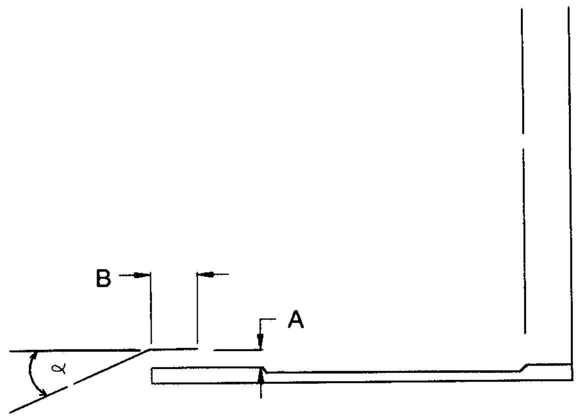

FIGURE 3.7 *Air foil design.*

Manufacturers do some silly things with this vital design parameter. One makes B too small to be really effective; another makes A too large and B too small and for rigidity puts a downward slanted return on the back edge of the foil, making it worthless for all practical purposes.

You should also consider the materials of construction of this front foil. One that is made of mild (regular) steel and painted with the same finish as the laboratory cabinets is a mess in a year or less. A stainless steel foil used in an acid environment takes a good beating because stainless steel just isn't stainless. While stainless steel can get pretty grubby after awhile, you can clean it up with some cleanser and elbow grease. By far the best is one made from mild steel and covered with a special coating that is specifically chosen for the procedures used in your laboratory. My experience tends to favor Kynar [2], however, pinhole-free Teflon [3] would be a very good choice. Teflon can be white where Kynar is a dark grey.

BACK BAFFLES

Referring back to the introduction, we said that the back baffle was the first major improvement to the alchemist's fireplace. Not only was it the first, but it may well prove to be the most important part of an operating hood (Figure 3.8).

As hoods are currently manufactured there are two slots (A and C) or three slots (A, B, and C) in the back baffle. A and C are normally adjustable

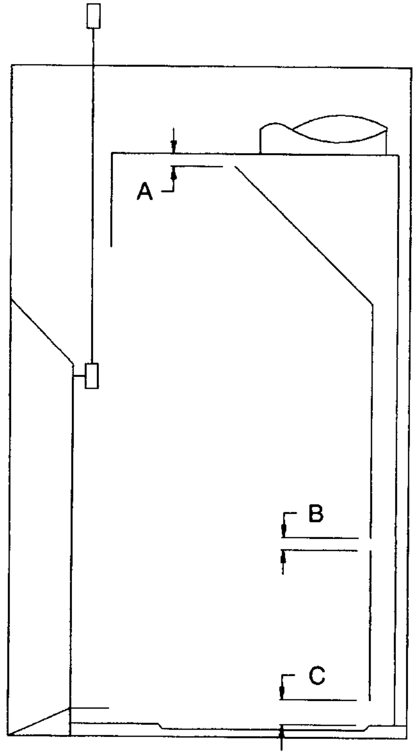

FIGURE 3.8 *Standard back baffle slot sizing.*

from a closed position to full open (roughly 2 in.). Slot B is usually a fixed size and varies from 1 to $1\frac{1}{2}$ in. Product catalogs state that these slots are adjustable to compensate for heavier than air and lighter-than-air gases. So much bunkum. Except for heat and natural gas it is rare indeed to utilize a lighter-than-air reactive gas. Most vapors, including solvents, are heavier-than-air. Procedures that utilize mercury or bromine produce a substantially heavier than air environment. Admittedly you have a great deal of dilution of reaction chemicals and products. But even so the average off-gas is slightly heavier than air even in a dilute state. Nonheated reactions are preordained to be more dense than their heated brethren.

Since we are primarily dealing with heavier than air products how do we adjust these "adjustable" baffles? Short answer: take the bottom "adjustable" part off and throw it away. The top opening should be set in the $\frac{1}{2}$-in. range and then screwed or glued in place so that it cannot be changed by an uninformed worker.

Not convinced? Okay, then let's see how a fume hood works in regard to various baffle slot settings (Figure 3.9) [4], [5].

Your principal exhaust action is past and over your work surface and the balance goes through the middle slot B and top slot A. With A open to $\frac{1}{2}$ in. you generate an area of swirl called the vortex. At this $\frac{1}{2}$ in. setting the vortex is confined to that volume in the top portion of the hood chamber that is behind and level with the bottom of the raised sash. Your work surface is well ventilated; we refer to this as "floor sweep" (Figure 3.10).

The top slot A receives the maximum exhaust potential since it is closest to the exhaust duct. When open it really diminishes the volume of air available to B, but most especially to C. The floor sweep that is essential

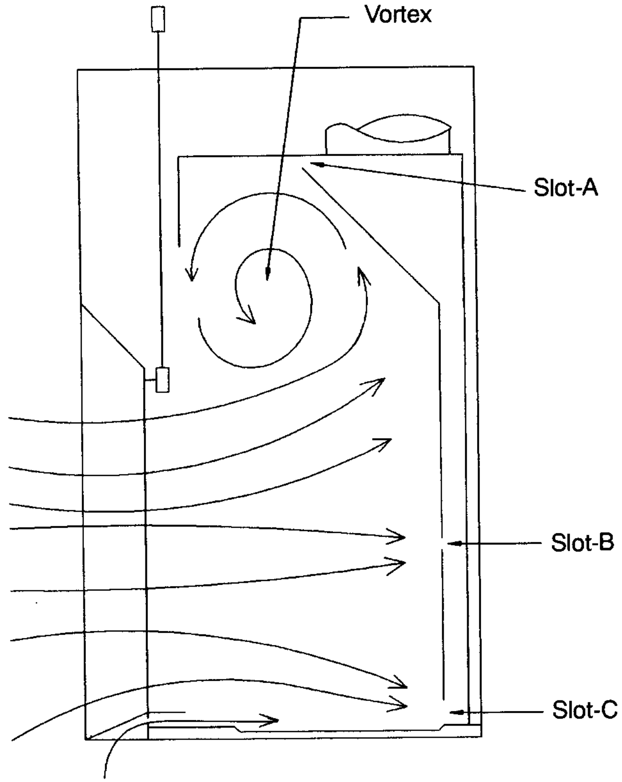

FIGURE 3.9 *Baffle slot settings. Conditions: A open $\frac{1}{2}$ in.; C wide open (2 in.).*

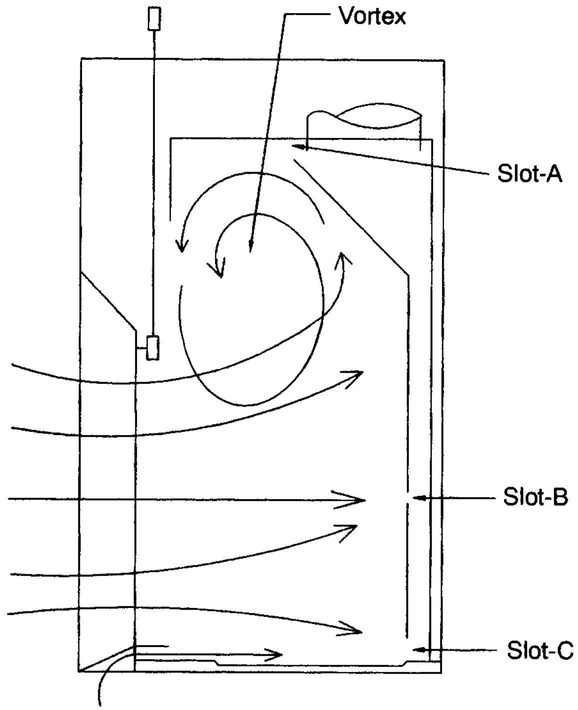

FIGURE 3.10 *Ventilation at work surface. Conditions: A wide open (2 in.); C wide open (2 in.).*

for safe fume hood operation is now marginal or worse. You concurrently have a substantial increase in the size and the amount of turbulence in the vortex area. Why is this? Well, the air that was hoping to be exhausted at A has acquired a fairly high velocity and speeds right past slot A. Due to the hood geometry it heads back down into the hood chamber where it is buffeted by the horizontally directed air just entering the front face of the hood at the bottom of the raised sash. hence the large swirl. With A wide open the vortex extends down below the bottom of the open sash, where it is subjected to more abuse.

The cross-drafts that are present in a laboratory regardless of the source can and do grab some of this interior hood air and force it into the laboratory proper.

The area located by the bottom of the raised sash is basically that of a square corner. This area also pulls air forward, and unfortunately this is about the right spot for a person's breathing zone. For this condition it is easy to see that the hood has become an unsafe area in which to work. It can, however, get worse, as we will see in Figure 3.11.

You now have no floor sweep in the hood. The size and degree of the turbulence of the vortex are now affecting the hood air pattern well down into the hood chamber. Any room disturbances, even those that would normally be acceptable for safe hood operation, totally disrupt the air exhaust pattern and allow copious quantities of hood-contaminated air to enter the laboratory.

What does this all add up to design-wise for the back baffle?

1. We do away with the adjustable slot closures at A and C.
2. We design A to be $\frac{1}{2}$ in.

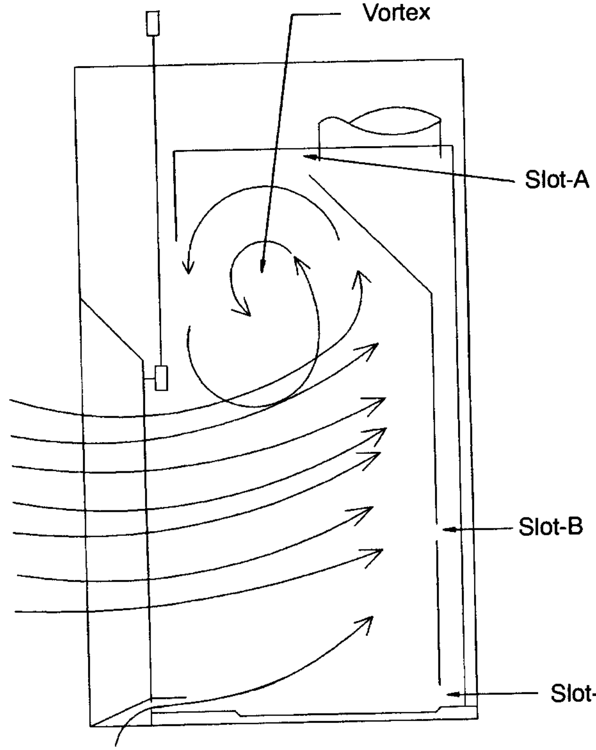

FIGURE 3.11 *Unsafe hood conditions: A wide open (2 in.); C closed.*

3. Slot B is fixed at $1\frac{1}{2}$ in. and located 14 in. above the work surface.
4. Slot C is a fixed slot 2 to $2\frac{1}{2}$ in. wide.
5. The clear air space behind the back baffle is 2 in.

While we are discussing the back baffle let us now consider another important design parameter (Figure 3.12).

Once slot A is set the body of the back baffle should be forward of the exhaust duct opening by at least 1 in., the reason being that the exhaust duct area has the greatest exhaust suction that is available to the hood.

If dimension X does not exist (no overlap), you have a much larger volume of air being exhausted in the center one-third of the hood as compared to the side wall areas. This cuts down on side wall ventilation and allows more turbulence to occur at these vertical leading edges.

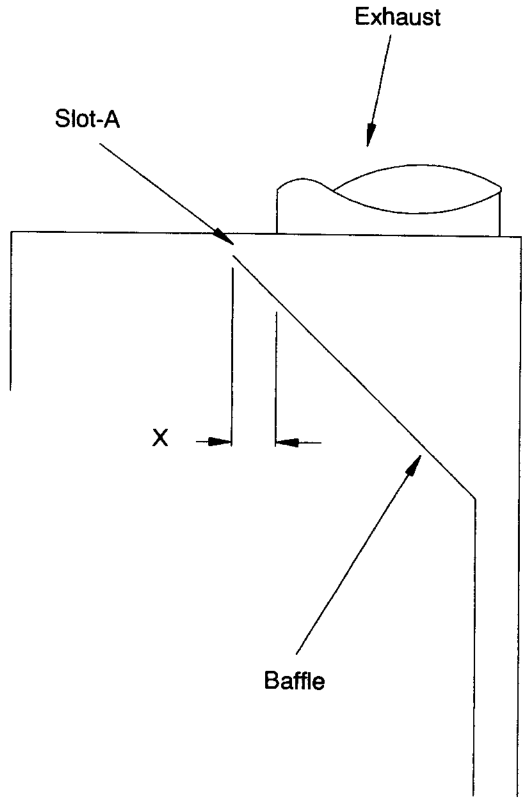

FIGURE 3.12 *Back baffle plenum design.*

In Chapter 9 we will be reviewing a large number of field tests of hoods (ASHRAE protocol). One of the first examples is the effect of slot openings on performance. If at this time the ASHRAE protocol is a "puzzlement" (àla "King and I"), be patient as we will also do a bare bones review of this important procedure.

We have reviewed the two major design areas of the hood superstructure: the air foil and back baffle. Now we move on to the sash and sash area.

VERTICAL SASH

The front sash offers an extension of safety that cannot be found on a full open-faced hood. If things start to go wrong and you need a quick place to hide, then the lowered sash offers this haven. It has other attributes, but for now we are going to be very general.

The sash is normally framed and fits into tracks in the hood side wall. It is counterbalanced to be reasonably easy to raise and lower and the glazing material is normally $\frac{1}{4}$ in. safety plate. The frame material can be stainless or painted steel. It makes little difference and depends on the user's preference and aesthetic requirements. The interior, or retaining part of the assembly, is another story. It should be a coated steel (Kynar or equivalent) and not painted or stainless steel. Rigid plastics compatible to your procedures are also very acceptable.

The two most annoying problems with a sash are (1) it is hard to move up and down and (2) it binds in the sash track.

The sash should be easy to move if the sash weights are of the proper size, and the pulley system is uncomplicated and well lubricated. You may not get, or need, ease of operation with a one finger push, sashes all have handles so you really are looking at a hand push/pull, not one-finger. It is more important to have the sash come down easily. If you have it set at a mid position, it should not creep either up or down.

The sash assembly fits into the track system with some type of spacer system. A well designed system is almost a guarantee of good operation. They should be simple and not require some type of field adjustments. If your sash binds due to the spacer/track alignment, it may not be the design but a poor installation of the hood in the laboratory. This is easy to check. With a measuring tape see if the distance between the side walls is the same at the top of the sash area as it is at the air foil area (plus or minus $\frac{1}{16}$ in.). If the hood opening is canted in or out down by the air foil, it isn't too hard to fix. Side walls will move a little if you persuade them with a hammer and a block of wood. You can most likely move them up to $\frac{1}{4}$ in. without causing any structural problems to the hood or the service piping. If you can't correct the problem, call the manufacturer. They would much prefer to have you as a friend than as a complaining user. But for heaven's sake call them when you first have the problem, not 10 years later.

HORIZONTAL SASH

A horizontal sash only moves from side to side, it has no vertical motion. The glass panels may or may not be framed. They are normally hung from a roller and track system and each panel overlaps its neighbors by at least $\frac{1}{2}$ in.

Horizontal sashes have suffered from poor reviews since they were first introduced and this has to be at least 75 years ago. The primary reason was that the original design was grossly deficient. The number of horizontal panels that were used were just too few. Let us use some examples to show what happened (Figures 3.13 to 3.15).

There is no sense in drawing any more sash configurations as it is now obvious that the number of sashes and tracks can make or break a horizontal sash system. You can have five sashes in three tracks, eight sashes

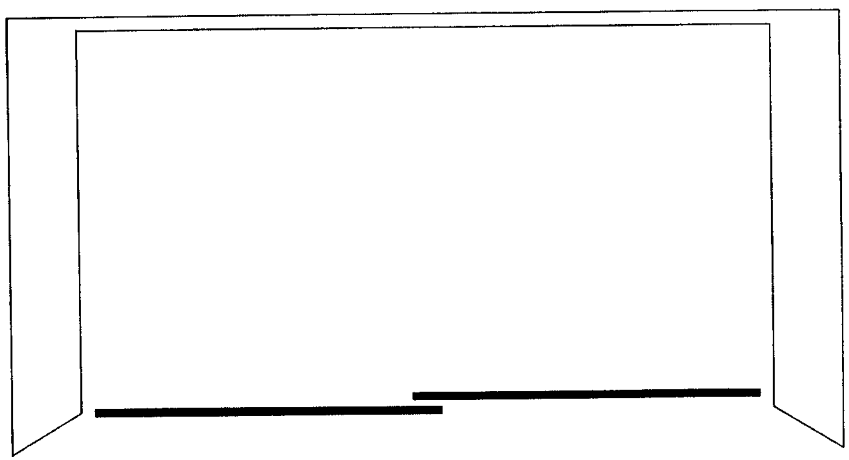

FIGURE 3.13 *Two sashes, two tracks. (1) You cannot reach the center area of the hood. (2) This barely provides a 50 percent maximum open sash area.*

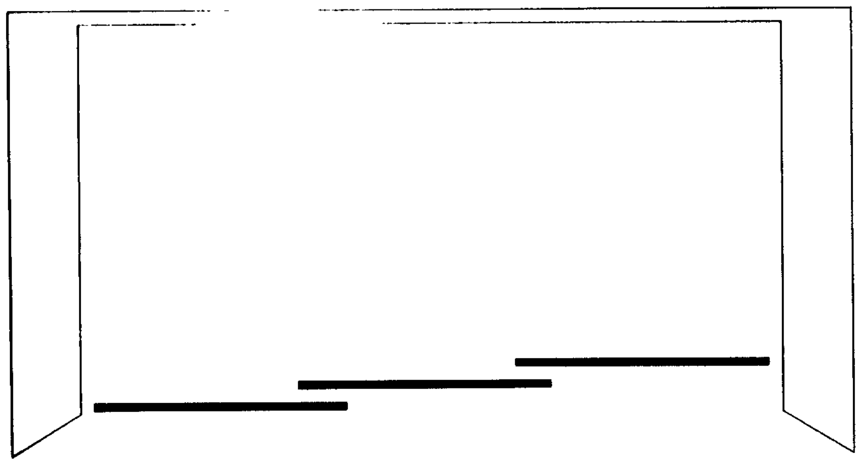

FIGURE 3.14 *Three sashes, three tracks. (1) You can reach the center area of the hood. (2) This provides 66 percent maximum open sash area.*

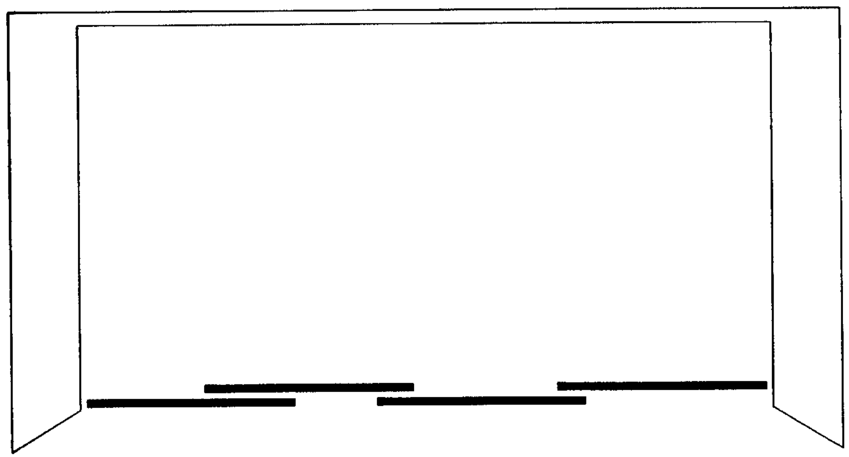

FIGURE 3.15 *Four sashes, two tracks. (1) You can reach the center area of the hood. (2) Panels could be made small enough (15 in.) to act as safety shieids. (3) This provides 50 percent maximum open sash area.*

in three tracks, or about anything you want depending on the width of the hood. Never make the sashes smaller than 14 or 15 in. as they just don't slide well. However, the 14- or 15-in. design allows the sashes to act as a series of movable safety shields. Over 15 in. is not comfortable for most people to reach around with both arms simultaneously.

A hood smaller than 6 ft should not be considered for a horizontal sash design. The opening is too small. Any size 6 ft and larger can work great; however, you must insist that the manufacturer provide a very strong track support and bracing system or you can have a lot of sag in the center, and the system will not work.

The primary reason for the use of the horizontal sash system is to effect a savings of the high-quality air (heated, cooled, and filtered) that you supply to the room and then exhaust through the hood. This air is very energy intensive and saving a substantial quantity of air is saving a lot of dollars. Not only can you save on laboratory operating costs on a continuing basis, but it also saves many dollars up front in reduced capital cost for mechanical equipment and it allows for smaller duct sizes. Each reason is good news to the mechanical engineer and to the vice president of finance.

To complete our evaluation of the horizontal versus vertical sash let us use our stick figures to do the talking (Figures 3.16 and 3.17).

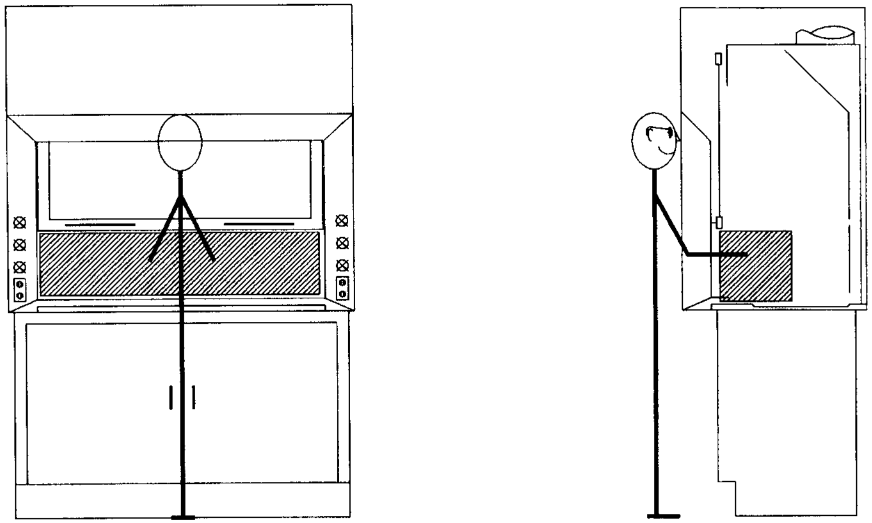

FIGURE 3.16 *Horizontal sash, 50 percent open. (1) Horizontally oriented access to hood. (2) Activity confined to cross-hatched area.*

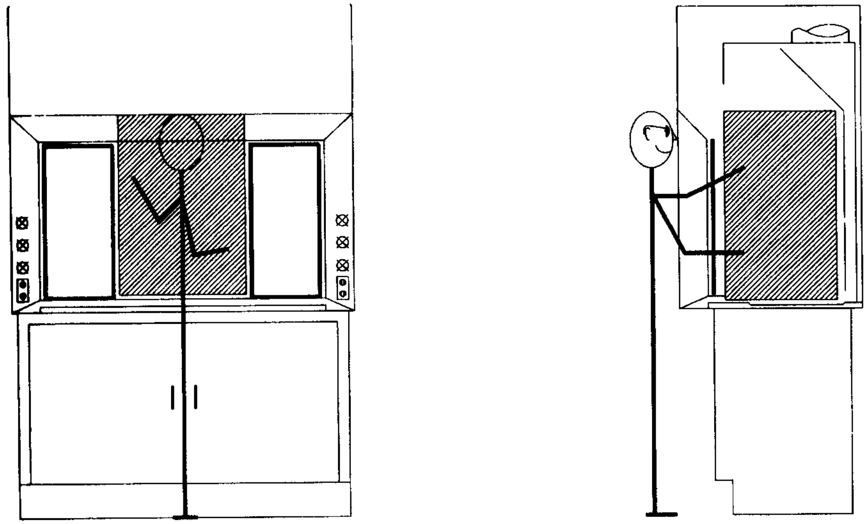

FIGURE 3.17 *Horizontal sash, 50 percent open. (1) Vertically oriented access to hood. (2) Activity confined to cross-hatched area.*

If your laboratory mission does not require any vertical reach and involves small work-top procedures, then a horizontal sash system could be an annoyance. If, however, you do need the vertical reach and you do want the savings incurred with a 50 percent reduction in air volume, then a horizontal sash system might just be for you.

If I am asking a laboratory complex to consider the horizontal sash system, I suggest that they have their maintenance shop make a mock-up for evaluation by their laboratory staff. They can use wood, aluminum channel, and plastic panels and make it so it fits, and is clamped, into the face opening of an existing hood. You will be amazed at how many converts, even supporters, you can make.

HORIZONTAL/VERTICAL

The combination of a horizontal system in a vertical rising sash is a hybrid that seems to be the answer to a laboratory group with multifaceted hood requirements.

In this system the vertical sash frame has the horizontal panel system built into the framing. They are normally unframed panels and are bottom supported. This costs a bit more for the hoods than the single purpose systems, but the payback is great and is a wise investment for diverse and changing laboratory assignments. The Monsanto Research Laboratories in St. Louis have installed this design, to the exclusion of all others, for the past 10 to 15 years.

SASH AREA

I use this connotation so we can explore other areas in or around the sash proper.

Where a vertical rising sash goes past the superstructure body, there is a space or gap built into the hood body so that the sash frame does not bang the hood lining every time it is raised (Figure 3.18). When the sash is in the up position, air enters the hood through this gap and is in turn exhausted through the duct system. This leakage amounts to roughly 10 percent of the total exhaust volume. If you want to save energy or increase a low face velocity (less than 60 fpm), then close this gap. It is easy to do this when the hood is built or even later in the field (Figure 3.19).

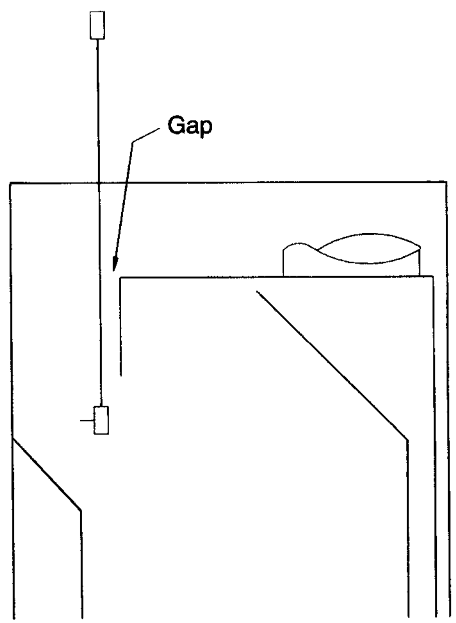

FIGURE 3.18 *Gap between hood body and sash.*

Purchase a sufficient length of 3- to 5-mil Teflon strip about 2 in. wide and fasten it to the top of the hood at the sash opening. The Teflon should ride on the glass by about $\frac{1}{8}$ in. to give a fair seal.

SASH STOPS

When using vertical sashes in an exhaust system where the mechanical engineer has only provided sufficient volume for a 50 percent sash open area, you might like to consider some sash stops. Whereas the design people depend on staff training to keep the sash at a maximum of half-open, we know that this discipline can disappear after a period of time. The sash stop is simple; it only stops the sash at a predetermined point when the sash is raised, not lowered. These stops are reminders of the half-closed (or open) philosophy, are not expensive and can be added to almost any framed sash hood at any time. They are better than nothing, and in many installations have proved quite successful (Figure 3.20).

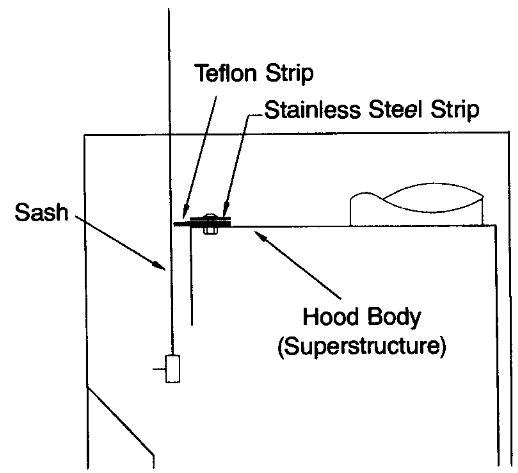

FIGURE 3.19 *Closing gap between hood-body and sash.*

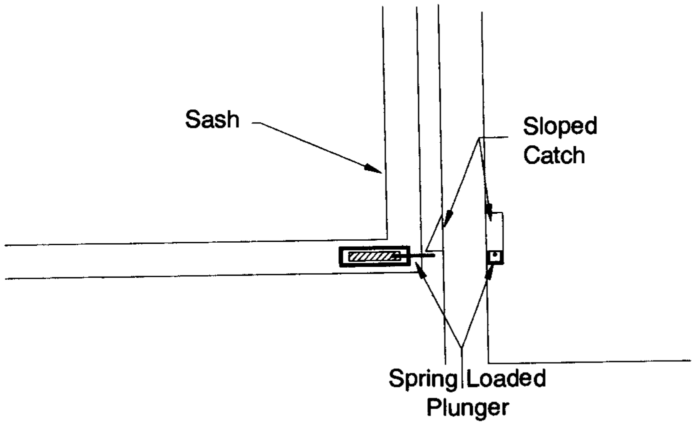

FIGURE 3.20 Sash stop.

BYPASS DESIGN

Horizontal and horizontal/vertical sash systems require some modifications to the bypass above the sash which is normally an unobstructed area. This type of bypass is used on all constant-volume fume hoods. When the sash is in the down position, the open area allows room air to enter the hood and maintains a reasonably constant exhaust volume.

Constant-volume hoods are NOT constant face velocity hoods; as the sash is raised from the closed position the velocity increases by a factor of from 3 to 4 depending on the clear open size of the bypass itself. The only so-called constant-velocity hoods are those equipped with variable air volume (VAV) control systems. Variable air volume systems are reviewed in Chapter 6.

Envision a horizontal/vertical combination sash with a conventional (full open) bypass area. Open one panel and then raise the sash. A large amount of air rushes into the hood by this path and not through the working sash area; the face velocity takes a real tumble. Again, you can rely on discipline to be sure all panels are closed when the sash is raised, but people can be forgetful.

This is easy to fix. Cover the bypass area with a perforated plate. Use the same material as the hood liner and with a drill or punch provide a 12 percent to 15 percent open area pattern. Why bother? If the sash is down and all the panels are closed, you still need some air ventilating the hood. That area under the air foil is too small to provide sufficient air at an acceptable velocity.

For a horizontal sash system the bypass area must also be fitted with a closure panel. Use the same material as the liner, only this time drill or punch a series of $\frac{1}{8}$-in. holes on 1-in. centers across a 2-in. central horizontal area of the closure panel. This too serves to give you upper hood ventilation when all the sash panels are closed.

MECHANICAL SERVICES

We will not delve into a lot of valve detail; several companies describe these products quite well in their catalogs. The hood manufacturers usually include only one brand of fixtures in their catalogs but can supply other brands and styles as made by any of the major fixture houses. Have your mechanical engineering people select the best valving system not only for you, but for the total laboratory complex. In an integrated facility it makes a lot of maintenance sense to settle on one valving system. Our aim is to use these systems to our utmost advantage, regardless of the brand.

The enclosed space in the hood side walls gave the mechanical people a chance to hide a lot of piping and wiring. It also provides a much better location, from the user's standpoint, to mount the service valves. The valve handles are up where you can see and reach them easily.

In the last few years some manufacturers have been producing hood valves where the valve body is mounted on the front surface of the vertical facia panel rather than being located in the enclosed space and being operated by a remote control rod. These are referred to as front-mounted valves; however, it is not a vital consideration in regard to hood performance.

Figure 3.21A shows a typical remote control rod valve assembly. If the valve starts to leak, it is hard to fix and the user has some diminished fine control over the valve settings. Figure 3.21B shows a front-mounted valve assembly with the valve stem and handle oriented straight out from the facia panel on the hood. Figure 3.21C shows a front-mounted valve assembly where the valve stem and handle assume the angle of the facia panel proper.

I am favorably impressed with the system shown in Figure 3.21C. The valve identification button is easier to see; such valves are easier to operate and control is better. The valve handle only protrudes about a $\frac{1}{2}$ in. past the

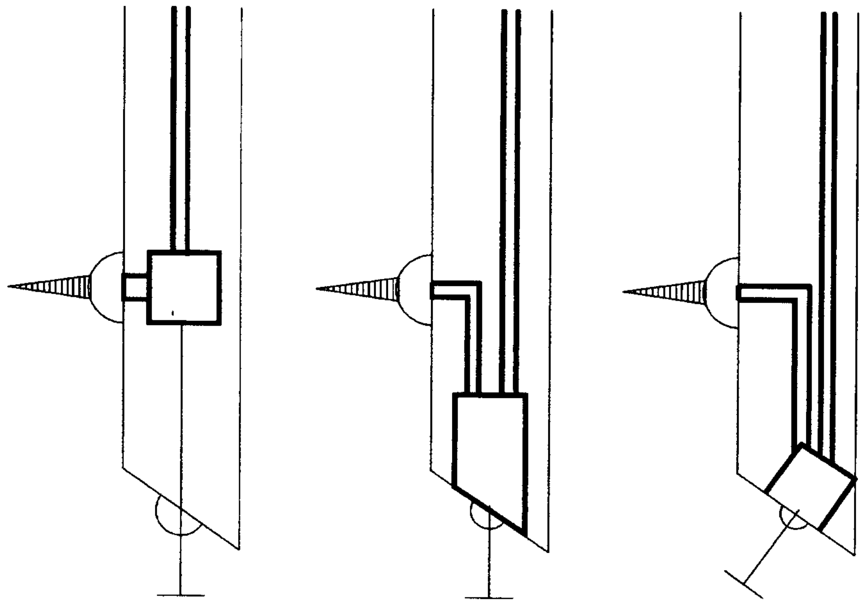

FIGURE 3.21 Hood valve assemblies.

front plane of the hood. Remote control rod valves and straight oriented valves stick out at least $2\frac{1}{2}$ in. With that much reach they are great at grabbing a lab coat.

The interior hose bibs and faucets should get some attention if only because of their finishes. Chrome-plated fixtures deteriorate with time. Those with an aluminum or bronze finish do much better and those with special epoxy finishes will out last the other two. You can get molded nylon and rigid PVC bibs and these can last a long time.

For hoods that exhaust fumes from perchloric acid digestions, commercial coatings are not compatible and chrome plating corrodes rapidly. Nylon is not acceptable; molded PVC is compatible and will last as long as the hood itself.

One last suggestion. Try to eliminate all metal that is not specially coated from the interior of general purpose hoods. This includes screws, nuts, and bolts. They rust and ruin the appearance of the hood and may present cross-contamination.

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4

Applied Product Design

In Chapter 3, Basic Hood Design, we defined the various parameters that govern good overall hood design and performance. While these criteria are dominant for all fume hoods, there are specifics of construction that are applicable to special purpose hoods. It is this area of "specialty" that we now explore so as to define and construct hoods for specific applications. We also review some of the "appliances" that can be, and in some cases must be, part and parcel of most hoods.

PERCHLORIC ACID HOODS

The perchloric acid hood is a single usage hood. It should only be used for perchloric acid digestions, and under no circumstances should it be considered for general purpose chemical procedures. Perchlorates formed during the digestion process are very unstable and when dry have a very rapid reaction rate that can cause violent explosions. Fortunately, these perchlorates are water soluble and we use this property to help contain the hazards of this chemical.

A perchloric acid fume hood, and its associated exhaust system, must meet these three basic parameters:

1. Made of materials resistant to reaction with perchloric acid.
2. Watertight construction to accommodate water wash down.
3. Void of sources of spark generation.

There are two materials that can be used for fabricating these hoods: type 316 stainless steel and rigid PVC.

Stainless steel is the material of choice by fume hood manufacturers since they have facilities to weld and polish this material. The resulting product is initially attractive and is structurally sound. Stainless steel is not really stainless, of course, when it comes to most acid applications and in a year or so the hood appearance is no longer pretty and shiny. It is still very safe to use, but pretty it is not.

Rigid PVC, on the other hand, is as attractive after a year of service as it was on day one. There is one major problem with securing hoods made of PVC; major hood manufacturers do not have the expertise or facilities to construct hoods from this material. The hood superstructure commonly is subcontracted to an outside firm with little or no knowledge of fume hoods but with expertise in plastic fabrication. Sometimes the results are marginal, sometimes great, and it all depends on the coordination between the hood manufacturer and the PVC fabricator. In some geographical areas there can be local fire codes that preclude the use of PVC, but it may be possible to secure a waiver. Have your hood supplier expend the effort to coordinate the engineering and construction with the outside fabricator, and you will be more than compensated for the effort. At the end of several years of service this material looks like new, whereas the stainless steel hood may have been replaced at least one time.

The water wash-down system, in the hood superstructure, must be designed and constructed so it provides complete coverage of the area behind the back baffle. It must do this without spraying water over the baffle system and onto the working surface of the hood. Some fabricators use spray nozzles, some perforated pipe; it has been my experience that a properly engineered and drilled pipe system can offer the best solution, (refer to Figure 4.1).

Remember that the ducting also needs an equally efficient wash-down system and here the spray nozzles do a very fine job. This ducting wash down system must include all the ducting from the hood to the blower and the exhaust stack from the blower to the environs. The blower does not need a separate wash-down nozzle because it is the recipient of the exhaust stack water. The blower needs a drain that is connected back into the ducting to the hood so that the water is properly handled.

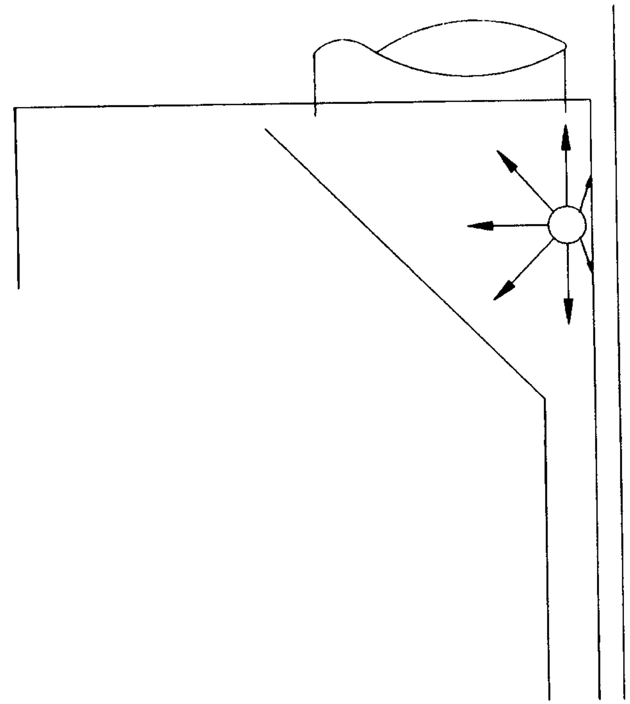

FIGURE 4.1 Spray-down pipe design for perchloric acid hood.

The proper design of these hoods provides a trough along the back of the work surface to accept the wash-down waters. Just be very careful in the design stage that the mechanical engineer does not provide more wash-down water than the drain can accept. If necessary the hood manufacturer will provide a larger drain connection to accommodate the required water volume.

The lighting fixture in these hoods is normally of explosion proof, incandescent design to take advantage of the waterproof and spark-proof features; however, it does not require that the wiring to the power source meet explosion-proof codes. All electrical outlets must be external to the hood work chamber and mounted in the hood vertical facia panels or some other outside surface.

Internal plumbing fittings such as serrated hose bibs, nozzles, and the like should be molded PVC. There are coated fixtures that are used by some suppliers, but these coatings are of an organic base and provide a source of hazard. Molded nylon is also available for plumbing fittings, but it too is combustible and provides a danger spot for perchloric acid oxidation.

The gasket material used to set the glass sash panel must also be perchloric acid resistant. Rigid PVC is normally used, although I did discover that one prominent, but now defunct, hood fabricator was using a sponge material that was not resistant to perchloric acid and did not know that it was unsafe. For your own safety please check with the manufacturer for the material being used.

Indicating manometers or low-flow alarms should be part of the hood system. Hot wire devices can be used if they have glass-enclosed sensors and operate on a 12-volt supply. When inserting a pressure probe for a manometer remember to weld the small tube into the ducting and orient it downward to at least a 45° angle. The probe piping should just barely extend into the duct so as to sense only static pressure and not both static and velocity pressure. Do not use a pitot tube, since it will indicate both velocity and static pressure and the small sensing holes can plug up rather rapidly. The connection from the ducting tube to the manometer should be standard copper or stainless tubing as it may be subjected to a very small amount of perchloric acid exposure.

The exhaust collar for the hood should extend at least $\frac{1}{4}$ in. down inside the hood chamber and be seam welded to the inside surface of the chamber. Refer Figure 4.2.

This offset allows the wash down water to flow directly into the hood. If you do not insist on this slight duct extension, the wash-down water will flow out of the ducting and spread laterally across the top of the hood and eventually will drip or flow unceremoniously onto the hood work area. It is amazing how few hood manufacturers recognize this as a required construction feature.

The frequency and duration of use for the wash-down system are an area that presents a lot of questions. Here are my suggested answers:

1. If only moderate amounts of perchloric acid are used, (100 ml or less) once each day should be sufficient.
2. If the quantity is close to 1 liter or more each day, the wash down could be done two or three times each working day at some convenient pause in the procedures.

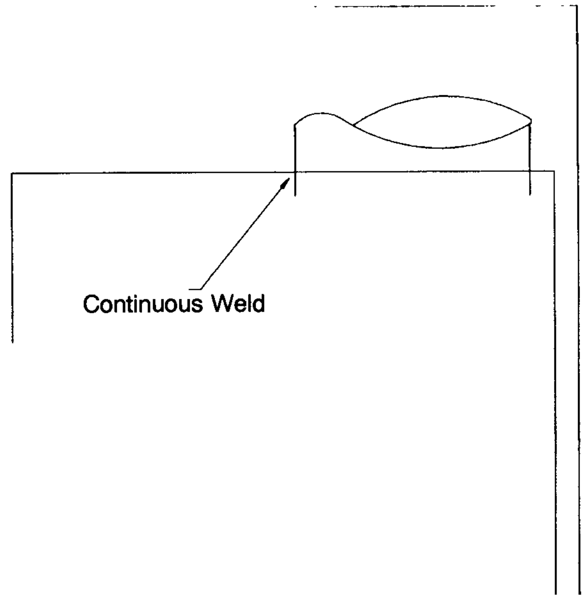

FIGURE 4.2 *Design criteria for welding and penetration of duct collar for perchloric acid hood.*

3. The wash down should be activated for a long enough period to ensure a thorough cleansing. This could be in the 5- to 15-minute range.

The wash-down system should be activated only with the exhaust blower OFF so that complete coverage can be achieved. Automated and timed wash-down systems are acceptable as long as they also turn off the blower during the water wash-down cycle. If the exhaust blower is on, you have air currents carrying the water out of the hood up the duct and away from the trough drain system built into the hood. The lack of the downflow of water also disrupts the full coverage needed to cleanse the contaminated areas.

Perchloric acid hoods do NOT fit into a central building exhaust system since they MUST have a dedicated duct and blower system. The ducting should be as short as possible and with few, if any, horizontal duct runs. Perchloric acid fume hoods are difficult, if not impossible, to locate on the lower floors of a multistory building.

I have not seen perchloric acid hoods integrated into a VAV control system. This does not mean that this is impossible; however, the numbers of this type of hood in use are quite small and the amount of air saved with VAV controls hardly seems worthy of the expenses involved.

RADIOISOTOPE FUME HOODS

The demands of the radioisotope hood have to do with two areas: cleanliness and load support for lead brick shielding. Type 304 stainless steel, cove-welded liners are most acceptable. Where the cove seams are welded they should be ground smooth and polished. The rest of the stainless liner should remain with the mill finish (referred to as a "2B finish"). The polished surfaces are shinier but much rougher than the mill finish. The work surface is normally supported by an industrial steel grating with a load-bearing capacity of 200 pounds per square foot.

Radioactive chemical hoods may require either or both particulate filters (HEPA) and activated charcoal (absorption) filters. There are reliable filter manufacturers that can supply both and will provide the proper sizes depending on the type of work being done and the volume of air to be treated. The building mechanical design engineer must be alerted to the use of these filters since their presence will require a substantial increase in available exhaust pressure for this hood system. Hood manufacturers fabricate housings for these filters; however, they are not nearly as good as those made by the filter companies. You will pay more for those supplied by the filter people but it is well worth the cost involved.

Do NOT equip a radioactive chemical hood with any type of wash-down system, as it will spread any contamination into your building plumbing system. These hoods are cleaned with wipes and mild cleansers, which are then disposed of through the solid radioactive waste program.

WALK-IN FUME HOODS

Simply put, a walk-in hood is a bench hood that has been extended down to the floor level and with some minor back baffle changes. These hoods are not intended to have the user walk in and out of the hood as it is being used. They are intended to allow you to walk in, set up some equipment, and then go back outside the hood before starting any active work. They are made to house large equipment and if they are operated with the sashes closed they can work quite well.

Generally, a walk-in hood with the front sash fully open does not perform very well. They are too susceptible to room air current and laboratory traffic pattern disturbances. They should only be used for active experimentation with the sash mostly or fully closed.

The adjustment of the back baffle slots is most critical so as to ensure an adequate floor sweep and minimum vortex formation. The top baffle slot MUST be set with a maximum opening of $\frac{1}{2}$ in. and the bottom slot MUST be full open (2 in to $2\frac{1}{2}$ in). Anything less and you can cross off the walk-in hood as a safety device.

The sash configuration for walk-in hoods can be either double-hung vertical rising panels or horizontal sliding panels. The horizontal sash configurations are inherently safer than the vertical rising design. For starters, they are easier to move since the vertical sashes require more complex track and pulley systems, longer cables, and usually heavier weights as compared to a bench hood. Since the horizontal sashes are easier to move, you are more likely to have them closed when using the hood.

The floor design of the hood can be a critical factor in walk-in hood performance. The room air currents at floor level are much different from those at eye level. Walking, air leakage under doors, and deflected currents from make-up air diffusers can cause disturbances that are greatly amplified since the room floor and the hood entrance are pretty much at the same level. This is one reason that hood floor sweep is so important to good operation. The use of the bottom front air foil is not as important since an entrance air roll is not generated as with a bench hood.

In installations where solvents are the major ingredient in hood protocol, I have installed a barrier to help keep solvent vapors from crawling out of the hoods and onto the laboratory floor. See Figure 4.3.

The block (A) presents a physical barrier to the vapor flow from the hood and the modified air foil minimizes the effect of the turbulence caused by the block. The two components can be made as an integral one-piece assembly out of welded stainless steel, sized to fill only half of the hood opening area and hinged from the side wall area of the hood. They are raised and lowered as the use demands.

One area you should be aware of is the floor drain. It is not unusual to use fairly large volumes of solvents in walk-in hoods. If the solvent container fractures or develops a leak, you could fill your drain system

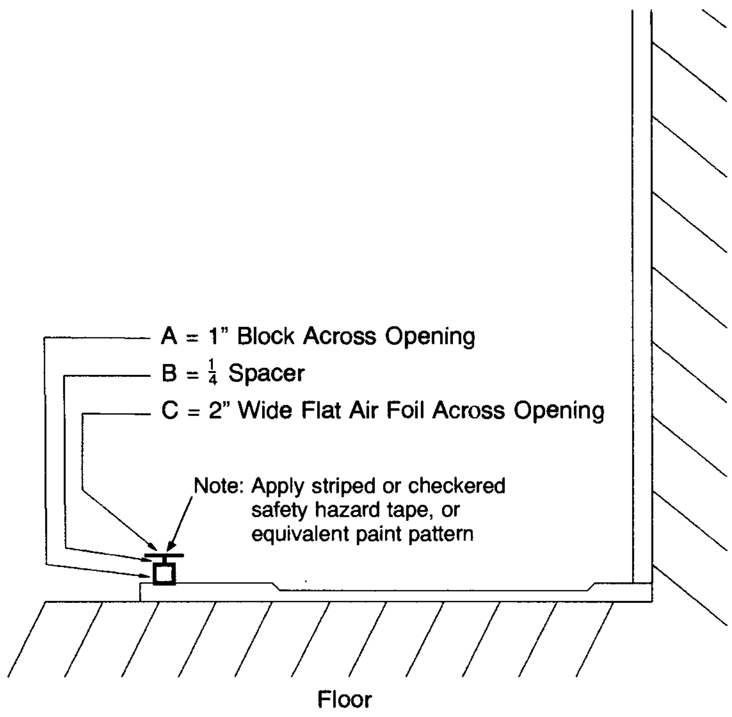

FIGURE 4.3 *Barrier control for walk-in hood.*

with a liquid that could be physically or environmentally harmful. You might have the manufacturer eliminate the drain altogether or make it so you can put in a standpipe overflow such as is used in sinks. In either case, the pan retainer design of the hood floor should be made deep enough to hold the volume of liquids that you anticipate using; the standard used by most manufacturers is for a $\frac{1}{2}$ in. indent. If you do design your pan deeper, greater than 1 in., then you probably will not need the barrier design of Figure 4.3.

CANOPY HOODS

Canopy hoods are meant to serve one primary purpose and that is to exhaust heat generated by a particular bench top group of equipment. If you have a series of ovens, it is well to exhaust this heat before it upsets your room temperature; drying ovens can also produce some unwanted odors that are not desirable to exhaust into the laboratory.

The canopy hood is not an efficient device from an air usage standpoint, but the volume of the exhaust air can be minimized. Locate the canopy hood as low as is practical for the equipment that is to be placed under it. Select a volume of exhaust air that would represent an average velocity across the opening of the canopy of from 50 to 75 fpm. This should be most adequate to remove all the generated heat and odors. Do not try to use a canopy hood as a general purpose type hood; it is not designed for this usage and will not safely contain chemical or toxic fumes.

DUCTLESS HOODS

In the past 4 or 5 years makers of the so called ductless hood have started to show a fairly substantial marketing effort. After all, a fume hood that does not require any exhaust ducting, has a built-in blower, does not upset any room air balancing, and can readily be installed by a couple of people has a lot of attractions. But like everything that is too good to be true, the ductless hood fits nicely into this category.

From a general chemical use standpoint, a hood must be able to withstand a wide variety of chemical abuses, elevated temperatures, noxious and toxic gases, and so on down a very long list of requirements. A hood system should not purposely add any type of contamination to the laboratory.

A profile of this type of hood indicates that the protection of the laboratory room from the activity in the ductless hood depends entirely on the retention capability of the particulate and absorption filters. Particulate filters (HEPA) are marvelous and have a retention efficiency for 0.3 micron (μm) sized particles of up to 99.99 percent, but they are not 100 percent efficient, and therefore some contaminants escape into the laboratory. HEPA filters do not withstand all modes of chemical attack, may not be heat resistant and can develop small leaks. The seal between the filter and the filter housing is critical for good performance. Of equal or greater importance, the activated charcoal filters that these units utilize have a minimal retention capacity at best. Absorption filters must truly be designed for the specific tasks at hand and may require flow rates and filter bed travel many times greater than are customarily available on these units.

For acceptable service the efficiency and capacity of both filter systems must constantly be monitored with a high degree of accuracy. Instrumentation to accomplish this costs many times that of this small hood itself.

The filter performance and capacity of the ductless hood are its Achilles' heel. If and when a 100 percent filter system is available then these units will become a viable laboratory hood; until then they could pose more of a hazard than a help.

GLOVED BOXES

During the 1940s and 1950s the use of localized containment systems for highly toxic and/or radioactive materials was pioneered by the University of California Radiation Laboratories in Berkeley. This rigorous containment philosophy was generally defined as concentrate, confine and control" (CCC).

The initial Berkeley gloved boxes were styled of ample size to perform a wide variety of chemical and physical procedures, but not macro chemistry involving large quantities of equipment or reagents. The chemist was isolated from the work by both the physical sides of the box and the long, almost shoulder length gloves that were sealed to the box and used to manipulate the work inside the unit. Refer to Figure 4.4.

A gloved box has its own small blower and filter system but is dis-

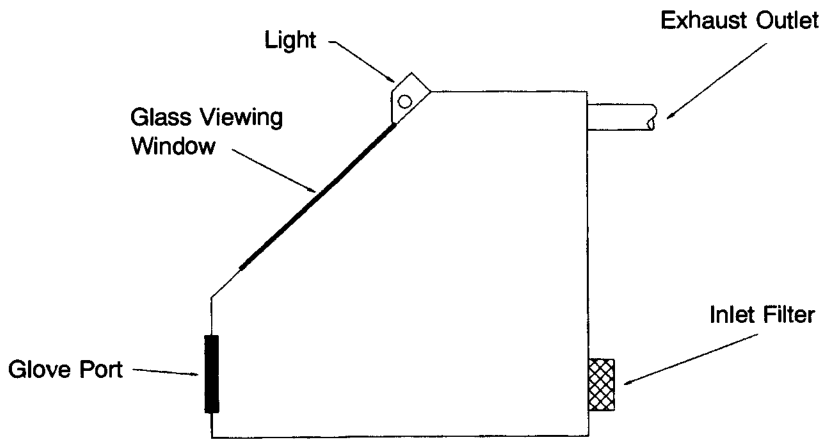

FIGURE 4.4 Gloved box.

charged into a central building exhaust and not back into the room. It has a slight negative pressure and the blower is in continuous operation.

While these small boxes did not lend themselves to macro manipulation, there was actually no limit as to the ultimate size as long as enough gloves were supplied to reach all of the interior equipment. I have seen some so called boxes that were all of 10 ft tall and 10 ft wide. but only four feet deep; gloved box gloves are nominally 27 in. long.

This "CCC" philosophy from Berkeley was very successfully applied to the production and/or manipulation of moisture- and oxygen-sensitive materials. There have been an untold number of very highly engineered electronic parts manufactured and special metallurgical procedures performed in this type of equipment.

The list of companies that manufacture this equipment has dwindled markedly in the past years. However, if you need help in locating a manufacturer or a systems designer refer to the *Buyers Guide* edition of some of the major trade journals.

LOCALIZED EXHAUST HOODS

Localized exhaust hoods fall into two distinct categories: One type is readily adjustable to bench top or distillation rack equipment so as to capture locally generated heat or off gases. In appearance they resemble an inverted funnel and are usually in the 10- to 12-in. diameter range. A flexible piece of tubing connects them to the building exhaust system. They are referred to by a variety of names but the most prevalent are "snorkel" or "elephant trunk." They work quite well as long as you do not expect them to do all things. Local exhausting they do, general exhaust procedures they do not do. All hood manufacturers provide these units complete with the connecting tubing, they are not expensive and are normally provided in painted steel or stainless steel.

The second type is most readily found in undergraduate chemistry laboratories. They are bench-mounted enclosures some 24 in. high and are 14 to 16 in. square, with an opening on the side that faces the student. The exhaust system is usually built up through the chemistry workbench so as to hide the ducting and give a better overall visual check for the teaching staff.

Regardless of the shape and size, they have some things in common: they really do not work well and due to their size can restrict many types of tall or wide apparatus. These undergrad hoods are mostly designed by a salesperson helped by the chemistry staff and assisted by someone with a degree in architecture. None of these people is likely to be well versed in the design of hoods and the results are unsatisfactory. When it is found that they do not provide acceptable containment, the usual response is to increase the exhaust rate, which is often more counterproductive than helpful.

This is not to say that a good localized hood for undergraduate chemistry cannot be designed, but to do so, someone will have to go back to basics, build one, test it, and go on from there. I have some suggestions where this designer can start and the direction to be taken.

Break up the turbulence that is caused by the square edges on all four sides and by the lack of a back baffle system. As a starting point I would suggest an air foil on each side, the top and the bottom. A simple back baffle system is then needed to provide both top and floor sweep in this small hood. All these suggestions are in keeping with the design parameters of Chapter 3. Refer to Figure 4.5.

I would suggest that the designers of this hood have a mock up made using any material that is handy and run some velocity and smoke tests. I feel that it will perform well with a face velocity in the 70-fpm range unless there are truly severe room cross drafts from the ventilation system or very active traffic patterns. I am certain that it will be a big improvement over what is currently being installed in undergraduate chemistry laboratories. When someone does design, test, and build a local ventilation hood that truly works, please write to me and the report will be included in a revised edition of this book 2 or 3 years down the road.

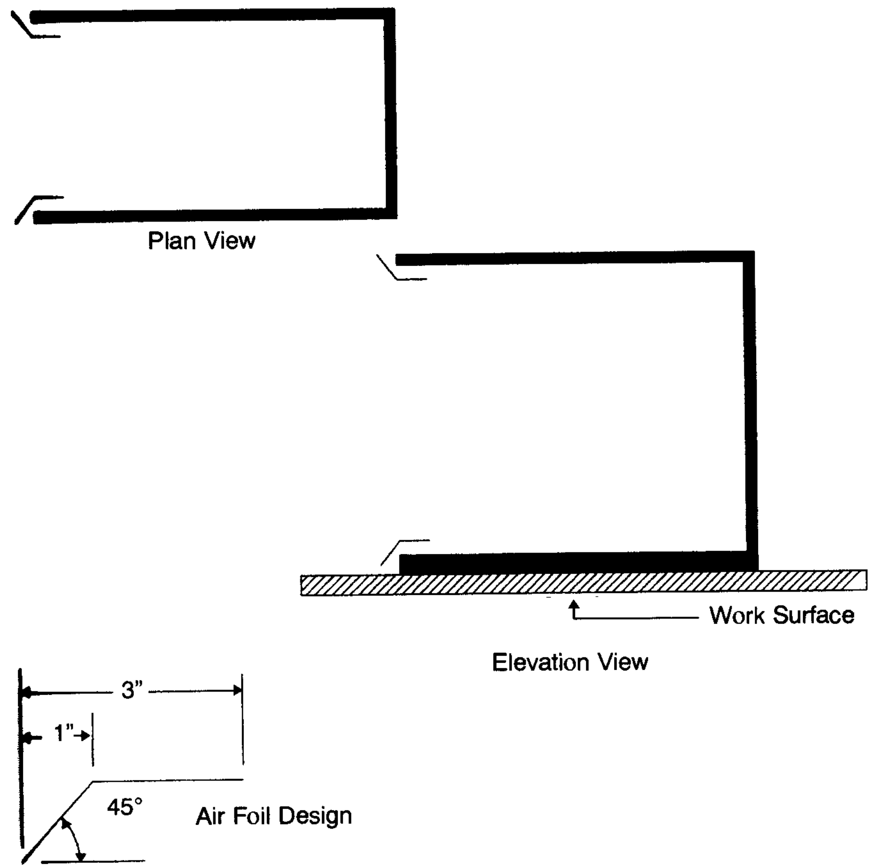

FIGURE 4.5 *Suggestions for design of undergraduate chemistry fume hood.*

SCRUBBER SYSTEMS FOR HOODS

There are reputable manufacturers of scrubber systems for exhaust systems of laboratory fume hoods and I will not invade their design territory.

There are specific areas that could benefit from a hood scrubber and this should be determined by an industrial hygienist after a proper amount of evaluation. I will not invade their territory either.

I only want to give some first-hand advice that I have gained from observing a number of scrubber installations that had been abandoned because they were not effective with the particular chemicals being used in the hood. First and foremost the lab people often did not understand that they are a piece of high-maintenance equipment and no one was prepared to service them properly.

During usage, scrubber systems need a regular clean-out schedule because they cake up and plug with the salts that form as the scrubber liquid evaporates during normal operation. This also dictates that the scrubber be located for ease of access. The water in the scrubber can freeze and usually does if the system is not located in a heated blower penthouse on the roof or top floor of the laboratory building. The de-mister system must be efficient so that you are not exhausting hail pellets during the winter and acid rain during the summer.

At a recent meeting of engineers I heard several remarks to the effect that almost all hoods need scrubbers to protect the environs. God forbid. I do hope that some environmentalist in local, state, or federal government does not decide to save the world by requiring scrubbers to be mounted on all hood exhaust systems. If we would review these units you would find that 50 percent were totally useless; 49 percent were not needed and only 1 percent were being effective. The costs involved in such a project would pale our national debt.

INDICATING MANOMETERS

For years I have been preaching the gospel that demands an indicating device on all fume hoods so that the user knows that it is drawing the proper amount of air. I felt vindicated in my belief when OSHA [1] included this little rule in their latest revision of the regulations for chemical laboratories.

When you tour new facilities, and even some older ones, you see a wide spectrum of instruments that are intended to show a hood operator the status of the exhaust air from his or her hood. These can range from very sophisticated digital read out instruments with color coded lights, simple dial or inclined manometers or perhaps just a piece of tissue taped to the bottom of the open sash.

For reliability, simplicity, and economy the indicating manometer is pretty hard to beat. They have hardly any moving parts, no electrical or electronic circuits, and they are simple to mount on the hood face and to connect to the exhaust duct. The manometer shows the hood user that there is a certain pressure drop when an adequate volume of air is moving through the hood and into the exhaust duct.

The manometer should be mounted at eye level on the face of the hood where the user can readily see it at all times. It is connected to the exhaust duct using standard laboratory flexible plastic tubing and a simple plate and metal tube configuration. This is screwed on the exterior of the ducting after a small hole has penetrated the duct proper. The metal tubing can be soldered or welded onto the metal plate depending on the material being used. Refer to Figure 4.6.

When the fume hood exhaust system is first put into operation and the face velocity is set to the proper (corporate designated) value, the pressure drop should be marked on the manometer sight glass using a marking pen

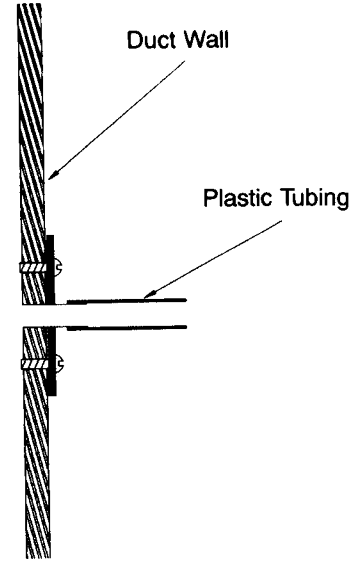

FIGURE 4.6 *Duct pressure tap for an indicating manometer.*

or a small brush with a quick drying paint. As long as the manometer needle is on or close to this line, the face velocity (exhaust volume) should be as it was originally established.

As time passes duct systems get dirty and may develop small (or large) holes, fan belts get frayed and slip, and the air volume will gradually decrease. The manometer will show the operator that this is happening by a drop in the indicated pressure and the building maintenance staff should be alerted. If the pressure decrease is very rapid, cease work in the hood, stop your experiments, and call the safety or industrial hygiene staff for immediate assistance.

The indicating manometer works only if the hood user pays attention to what it is indicating. Pay little or no attention to it, and it is useless.

ALARM SYSTEMS

Since I am pounding the drum for the manometer family I will state up front that you can very easily have an audible and visual alarm system using the manometer by adding electrical and/or electronic circuitry. Some manometers are manufactured with adjustable alarm set points as integral parts of the instrument. Other manometers can have adjustable subassemblies added to establish an alarm set-point system. To see what is available I suggest that you contact Dwyer Instruments, Inc., P.O. Box 373, Michigan City, Indiana 46360 for a catalog; they are the principal manufacturer of this equipment. You can go well beyond the simple dial manometer as such; there are digital read-out units, those with sequencing LED light patterns, and I am sure there are new products that I have never seen. Review them all and select the type that is best for you.

Other types of low-flow alarms, not based on static pressure drops, are also readily available. You will find hot wire sensors that are easily mounted in the hood side walls. There are other very sensitive instruments that sense the difference in pressure between the room and the hood chamber and translate this into face velocity. VAV systems, if you are fortunate enough to have one as part of your hood control instrumentation, are all designed to indicate and alarm for a low-flow situation.

As time passes I am sure that newer and better systems will be developed by the instrumentation fraternity. I would only ask that these companies determine what values should be indicated by their equipment. Currently, there is one company that tells the user that the air flow is in the energy efficient range, whatever that may mean. I want to know that my hood face velocity is safe, not energy-efficient. A little cooperation between the manufacturer and the user would be a great help to all.

A large number of alarm systems allow for the set point to be adjusted by the user. This is a very bad idea. The laboratory worker should know that his or her hood has an adequate face velocity. But the alarm set points should be determined and set by the safety staff. A researcher has enough to think about without getting involved with hood face velocity and pressure drop settings; both of which they probably do not fully understand. All alarms should be both visual and audible. The audible sound should have a "locked-in-position" silencing switch and only the supervisor should have the key; the visual component should remain lit until the low-flow situation is corrected.

All laboratory personnel from the janitor to the director of research must be aware that when a malfunction occurs it must be reported to the safety group immediately. Once a hood is not safe due to the lack of sufficient air exhaust, it must be closed to any continuing procedures until the air flow can be properly restored. If the failure occurs during the critical part of an experiment or process, some hard decisions must be made and people alerted. DO NOT TURN OFF THE AUDIBLE ALARM AND CONTINUE WORKING. You could very well endanger yourself and all the other occupants of the room. The low-flow alarm says the hood is unsafe to use. Heed the warning.

MATERIALS OF CONSTRUCTION

The principal item that allows for a large choice of materials is the hood liner. The materials range from stainless steel to cement board to various plastics. Each can have a special place for hood liners and new materials are being presented all the time. The hood manufacturers do not produce these lining materials; they buy them and cut them to the proper size for their products. This gives each manufacturer access to the same materials as a competitor. This means that the hood customer is not denied any type of material because of the choice of the company that might make the hoods.

To select the proper lining material for your usage, have the hood fabricator supply you with the specification documents for all the various materials; this will include both the chemical resistance schemes and the physical properties. It is wise to select a material that provides you with a white, or at least a very light (color-wise), interior hood finish. Be certain that the finish is resistant to your reagent usage and preferably is not too glossy. Hood chambers that are dull and dark are really depressing and become difficult to face each working day. The material should be reasonably smooth and easy to clean. If it is a painted finish, it should be capable of touch up or repainting in-place.

To minimize rusting problems it is wise to eliminate as much exposed metal from the interior of general purpose hoods as is possible. Plastics, if compatible, are good choices. Special coatings over steel are equally as good as long as the finish is extremely resistant to both chemical attack and physical abuse. I mentioned earlier in Chapter 3 that I had a preference for Kynar coatings. There are or will be others and it is wise to keep testing proposed finishes as they come on the market. Hood manufacturers can provide a lot of data for you but they have to be pushed and prodded to provide you with the latest technical data and the associated cost of the coating process. The usual ploy by fume hood salespeople is to talk you into the use of stainless steel in lieu of special coatings and/or plastics. If this happens to you, choose a new supplier or at least another salesperson because this type of presentation shows a low level of expertise. Remember what we said earlier: "Stainless ain't."

REFERENCE

1. 29 CFR 1910.1450, U.S. Government Printing Office, Washington DC, (1990).

5 Face Velocity

For impact, I am going to repeat part of the Introduction of this book with some of the historical events that have shaped today's understanding of fume hood face velocities.

Fume hood face velocity has been a very lively and controversial subject for health professionals and engineers since the 1930s and 1940s. In the early days of the Atomic Energy Commission (AEC) a hood face velocity of 50 fpm was considered adequate and safe. These were also the years of a widely and wildly expanding hood population.

Advanced hood design was just beginning in the 1940s and many questions were arising as to what was a safe face velocity. Unfortunately,the "more is better" fraternity became very vocal and convinced the majority of fume hood users that much higher face velocities were safer. At that time air velocity was the only parameter that could be measured objectively. There was some use of smoke wands and smoke bombs, but the results were more subjective than objective, and the face velocity was the primary focus of so called hood safety.

As the 1950s, 1960s and 1970s came and went, these "experts" were increasing the face velocity as rapidly as possible. From 50 fpm they had gone to 75, 100, and finally to 150 fpm and higher before some calmer heads prevailed. Some members of the fume hood community recognized that the increased velocities had not fostered safety and were driving the cost of building and operating laboratory buildings out of reach. In the late 1970s the American Society of Heating, Ventilating and Air Conditioning Engineers (ASHRAE) instituted a monumental research project [1]. Dr. G. W. Knutson and K. J. Caplan, undertook a project to establish a method of quantitatively testing fume hoods. We will study their report and resulting protocol in Chapter 9, but for now let us concentrate on a cursory overview:

1. Face velocities in the 60 to 100 fpm range provided acceptable and safe hood operating conditions.
2. Room air patterns can account for at least 50 percent of unsafe hood performance.

These data were reviewed and then published in 1982 by the American Conference of Governmental Industrial Hygienists (ACGIH) [2]. For the first time there existed a defined statement regarding hood safety and face velocities that was promoted by a society of health professionals and substantiated by excellent research.

It is only now (1993) that this information is being widely disseminated. Some manufacturers still fall into the trap of the 1960s and 1970s and rely on face velocity as a safety criteria rather than hood and room ventilation design. Some manufacturers' catalogs still reflect parameters as outlined in SAMA Standard LF 10-1980 [3], where so called class A, B, and C hoods were represented with face velocities up to 150 fpm. In view of more current data, these A, B, and C classifications and their implied face velocities should totally be ignored.

Set your hood face velocity based on the latest data and on factual rather than emotional presentations. I sat in a meeting with the design staff of a very large corporation and the staff industrial hygienist made a great statement: "We work with nasties so we will use 150 fpm as the design velocity." It is difficult to assess what that decision cost the company in increased costs and decreased safety. Whatever velocity you choose is up to you, and you must defend it to both the laboratory and the accounting staff. If you have chosen more than 100 fpm, go back and reevaluate your decision, or more likely the decision of an aging vice president or director of research.

To effect savings of the exhaust air, and thereby monies, many facilities are adopting policies that the fume hoods should be used with the front sash in a partially closed position and the volume (velocity) established with this criterion in place. This is not all bad but it sure can present a bag of worms if all the facets of the policy are not examined and recognized.

Regardless of the checks and balances used to require that the hood is actually used at the partially closed (or open) position, they will fail at least 25 percent of the time, and this figure may be as high as 50 percent depending on the integrity and reliability of user discipline. Laboratory personnel are not used to working with a partially closed sash and it will take both time and training to ensure compliance.

To be on the safe side, the measured face velocity of the hood, at a partial opening, should be close to what the velocity should be with the sash fully open. Remember the ACGIH set 60 fpm as an accepted minimum. As an example, if we have a hood with a 30-in. available sash opening, but want to use 18 in. as the restricted size opening then the ratio of 18 to 30 in. is 60 percent so the velocity at the 18-in. opening should be 60 fpm/60% or 100 fpm. If you are not comfortable with a minimum velocity of 60 fpm, you will have to insert a value that you prefer. Remember that overall sash opening height varies with manufacturer and with model, so you have to accommodate whatever the overall size actually is.

One last comment: Let us say that you choose 100 fpm as your minimum value; then when the sash is partially closed the measured velocity will be close to 170 fpm. Someone says it is too high because they have read that face velocities above 150 fpm can in themselves be detrimental to hood performance. The hazardous 150 fpm that they are referring to is the one present when the sash is full open. Except for sucking an occasional wipe up and out of the hood, these higher velocities become less and less of a disruptive factor, regarding operating performance, when the sash is moved to a smaller partially closed position. We discuss partially closed sash operation in Chapter 7.

REFERENCES

1. K. J. Caplan and G. W. Knutson, "Laboratory Fume Hoods, a Performance Test," RP 70, *ASHRAE Trans.*, 84, (I)(1978).
2. *Industrial Ventilation Manual*, 17th edition, American Conference of Governmental Hygienists, Cincinnati, 1982.
3. Laboratory Fume Hoods, SAMA Standard LF 10-1980, Scientific Apparatus Makers Association, Washington DC, (1980).